ACHELION CAPITAL MANAGEMENT, LLC

ARMS FSD

Full Autonomous Operations — Master Build Document

For: Josh (Lead Developer)

Version: 1.1 — March 24, 2026 | Revised: Immediate-Build Frontiers Added

Classification: CONFIDENTIAL — GP and Development Use Only

v1.1 adds Section 11: Five frontiers — what gets built now vs. later. Three frontiers are fully buildable today at zero incremental cost. Josh starts with Section 11 before anything else.

"Silence is trust in the architecture."

0. The Mandate

The PM declares intent. ARMS executes everything else.

This document is the complete engineering specification for transitioning ARMS from a dashboard-and-human-execution system to a fully autonomous decision and execution architecture. Every module described in PM Playbook v4.0 and THB v4.0 must be connected to a live execution pipeline by the end of Phase 3.

v1.1 adds Section 11 — the Immediate-Build Frontier specifications. Three of the five long-term frontiers are fully buildable today using free or low-cost tools with zero rearchitecting required when expensive upgrades arrive. Josh reads Section 11 before starting Phase 1.

The governing principle: human involvement at the execution layer is not a safety feature. It is latency and error risk. The only legitimate human touchpoints are thesis seeding via the SENTINEL protocol, regulated confirmation windows that exist to satisfy fiduciary law, and GP-level strategic override with documented justification and co-sign. Every other decision is owned by the system.

1. System Architecture Overview

ARMS FSD is a seven-layer autonomous system. Each layer produces typed outputs consumed by the next. No layer polls for human input except at designated confirmation windows. The system is a pipeline, not a collection of scripts. Build it that way.

Layer	Name	Function	Outputs to
L1	Data Ingestion	Market prices, macro signals, VIX, yields, options flow. Feed-agnostic plugin architecture — free feeds now, institutional feeds later.	L2 — Macro Compass
L2	Macro Compass	Regime scoring composite. Outputs current regime + RPE probability distribution of next transition.	L3 — ARAS
L3	ARAS + v4.0 + v5.0 Modules	FEM, PDS, PTRH, CDF, VARES, LAEP, SSL, MICS, RPE, CCM, TRP, AUP, SLA run after portfolio snapshot.	L4 — Master Engine
L4	Master Engine	Portfolio construction. Applies ceiling, MICS-derived C^2 weighting, CCM calibration factor, decay.	L5 — Order Book
L5	Order Book + LAEP	Converts allocation delta to executable orders. VWAP/market mode, priority, slippage budget.	L6 — Execution
L6	Execution API	Submits orders to broker. Confirms fills. Logs all executions with triggering module and rationale.	L7 — Audit + Monitor
L7	Audit + Daily Monitor	Immutable session log. Daily monitor 6AM CT. Session Log Analytics monthly. LP narrative engine reads from log.	PM / LPs

2. Four-Tier Autonomy Framework

Every decision in ARMS is classified into one of four tiers. Tier determines whether the system acts immediately, waits for a veto window, requires confirmation, or defers entirely to human judgment. No decision may be reclassified upward without GP documentation.

Tier 0 — Fully Autonomous

Zero human touch. System executes immediately. No override accepted.

Action	Trigger	Execution
Regime ceiling enforcement (ARAS)	Composite crosses regime threshold	Immediate
PDS DELEVERAGE_1 (−12% HWM)	NAV falls 12% from high-water mark	Immediate — 60% ceiling
PDS DELEVERAGE_2 (−18% HWM)	NAV falls 18% from high-water mark	Immediate — 30% ceiling
PTRH roll	DTE reaches 30 days on any QQQ put	Same session
PTRH regime resize	Regime transition confirmed by ARAS	Immediate
SLOF activation / deactivation	RISK_ON/WATCH: on. NEUTRAL: reduce. DEFENSIVE/CRASH: off	Immediate
CDF weight decay (0.80x at 45d, 0.60x at 90d)	Position underperforms QQQ by 10pp over 45 consecutive days	Next open
LAEP mode switch	VIX crosses 25 or 45	Immediate
MICS scoring on position initiation	Every new position through SENTINEL	Auto — no PM input
RPE update	Every 5 minutes with data refresh	Continuous
Session Log Analytics	Monthly after log compilation	1st of month
SSL daily computation + worst-scenario alert	Daily after portfolio snapshot	Pre-market 5:30AM
Systematic Scan Engine weekly run	Every Monday pre-market	6AM CT Monday
Daily monitor generation + delivery	Scheduled daily	6:00AM CT

Tier 1 — Default Execute (PM Veto Window)

System executes unless PM vetoes within defined window. No response = execute. Default is action, not inaction.

Action	Trigger	Veto Window
PERM covered call execution	Unrealized gain exceeds 30% on any position	4 hours
ARES re-entry T1 deployment	All 4 ARES gates clear + VARES sizes tranche	2 hours
CDF Day 135 orderly exit	Day count hits 135 without conviction reaffirmation	24 hours
FEM ALERT trim recommendation	ALERT (>80%) persists 24h without PM reaffirmation	24 hours
SENTINEL-approved position initiation	All 6 gates pass + regime NEUTRAL or better	4 hours
Thesis Retirement Protocol execution	180d in book + CDF 0.60x + bear evidence present	24 hours

NOTE: Confirmation queue must be built with three response options from day one — 'no new information, execute' (immediate), 'new information present, hold' (opens source declaration form), 'veto with rationale' (written reason required). Time-based windows are the fallback, not the primary interface. See Section 11.5.

Tier 2 — Confirm Required + GP Co-sign

System recommends. PM confirms. GP co-sign required. No default execution.

Action	Why Human Required	Window
ARAS regime override	PM holds pre-consensus insight the model cannot encode. Logged permanently.	GP co-sign
Architecture AB structural change	Changing 58/20/14/8 requires thesis-level conviction about civilizational shift weighting.	Full GP vote
LP communication narrative approval	Legal and relationship context. Auto-generated; PM approves before send.	2 hours
MICS override by ±2 or more levels	Extraordinary circumstance. PM must document specific rationale.	GP co-sign

Tier 3 — Human Only

No system recommendation possible. This is where human edge lives and ends.

Thesis seeding	Pre-consensus pattern recognition from lived experience. What is known before it enters any dataset.
LP relationship management	Trust, credibility, and emotional intelligence in capital conversations.
GP strategic vision	What Achelion is, what it stands for, where it goes.

3. SENTINEL v2.0 — New Position Protocol

SENTINEL v2.0 introduces three structural upgrades: a quantitative mispricing score at Gate 3, a mandatory adversarial cross-examination layer, and a source quality taxonomy at Gate 6. The output of Gates 1–5 feeds directly into the MICS formula — the PM no longer assigns a free conviction level.

#	Evaluator	Gate Question	Standard	Fail Condition
1	AI	**Civilizational shift — not a product cycle?**	Binary. The test: does this company exist whether or not any individual product succeeds? No hedging. No partial credit.	Fail = rejected. No appeal.
2	AI	**Non-optional — cannot be routed around?**	Binary with AI research. Models substitutability and structural necessity. 'Potential' does not pass. Current necessity only.	Fail = rejected.
3	AI + QUANT	**Quantitative mispricing score (0–30)**	Three factors 0–10 each: multiple mispricing, institutional positioning gap, consensus framing gap. Cat A/B: threshold 14/30. Cat C: threshold 20/30. Narrative cannot override the number.	Below threshold = fail. No override.
4	AI + FEM	**Architecture AB fit — no concentration without paired trim?**	FEM impact modeled before entry. WATCH→ALERT: paired trim required. Already ALERT: blocked. Mechanical gate.	ALERT worsened = blocked.
5	AUTO	**Regime timing — entry defensible?**	RISK_ON/WATCH: proceed. NEUTRAL: VARES sizing. DEFENSIVE: queued. CRASH: blocked. No override.	DEFENSIVE/CRASH = queued 90 days.
6	PM SOURCE	**Pre-consensus source declaration**	Source category declared from taxonomy (Section 4). Feeds MICS formula as source_score. No response = Cat 'no PM input' score (4/10). Cat D not accepted.	Cat D = rejected.

4. Gate 6 Source Quality Taxonomy

The source taxonomy is the mechanism by which free conviction assignment is removed. The PM declares a category. The category feeds the MICS source_score. Cat D does not exist as an input field — the system will not accept it.

Category	Definition	Examples	MICS source_score	Gate 3 threshold
Cat A Primary source	Direct conversation with founder, engineer, customer, or supply chain participant with firsthand knowledge not yet public.	TSMC supply chain VP describing capacity allocation. Hyperscaler engineer on internal tooling. Founder explaining structural advantage before public announcement.	**10 / 10**	**14 / 30**
Cat B Pattern recognition	PM has seen this exact configuration before in a prior cycle. Structural analog is clear. Pattern not yet in consensus.	NVDA 2020 rhymes with Cisco 1996. TSLA energy/AI thesis before auto framing dissolved. 28 years of cycle recognition applied to current setup.	**8 / 10**	**14 / 30**
Cat C Synthesis	Synthesis of public information not yet at consensus framing. PM connected dots the market has not yet connected.	Five earnings call transcripts revealing a supply chain shift no analyst has framed. Macro data connected to sector implication absent from Street research.	**5 / 10**	**20 / 30**
No PM input (silence)	PM has no pre-consensus insight to add beyond what the system already knows. Silence is trust in the architecture. System executes at MICS-implied sizing.	4-hour veto window expires with no PM response. MICS uses source_score of 4/10 — system proceeds at reduced conviction relative to Cat A/B. This is a valid and expected outcome.	**4 / 10**	**Gate 3 threshold still applies**
Cat D Does not exist	Strong opinion. Gut feeling. Conviction without identifiable source. Not a category. Not an input field. The steering wheel we removed.	Every rationalization a PM has made while overriding a risk system. The entire history of human error in portfolio management compressed into one category the system will not accept.	**REJECTED**	**REJECTED**

5. Five New Modules — Sharpening the Weapon

These five modules extend ARMS into anticipation, self-calibration, symmetric upside capture, proactive LP trust-building, and thesis retirement. Build status for each is specified in Section 11.

5.1 Regime Probability Engine (RPE)

From reactive to anticipatory. Pre-position before transitions rather than react to them.

File	src/engine/regime_probability.py
Runs	Every 5 minutes with data ingestion cycle
Build status	Phase 1 — see Section 11.1
Feeds into	ARAS advisory, PTRH sizing, SLOF margin, cash accumulation

5.2 Conviction Calibration Module (CCM)

Self-correcting conviction scoring. The PM who calls it right gets amplified. Noise gets corrected.

File	src/engine/conviction_calibration.py
Runs	Quarterly after session log compilation
Build status	Phase 2 — requires 90 days of session log data
Feeds into	Master Engine conviction_weighting_factor

5.3 Thesis Retirement Protocol (TRP)

Exit with the same rigor SENTINEL applies to entry.

File	src/engine/thesis_retirement.py
Trigger	180d in book + CDF at 0.60x + bear case evidence present in current data
Build status	Phase 2
Feeds into	PERM exit queue (Tier 1 default-execute)

5.4 Proactive Intelligence Digest (PID)

Transparency as competitive advantage. The LP who sees the system working every month never redeems.

File	src/reporting/proactive_digest.py
Frequency	Monthly — automated
Build status	Phase 3
Contents	Current regime + scoring · Top 3 SENTINEL analyses · SSL worst-scenario vs. prior month

5.5 Asymmetric Upside Protocol (AUP)

ARMS is currently built to protect capital. AUP makes it equally precise at compounding it.

File	src/engine/asymmetric_upside.py
Trigger	All 5 conditions: RISK_ON + avg conviction >7.5 + FEM clean + RPE <20% WATCH prob + SSL net loss <12%
Build status	Phase 2
Output	Tier 1 SLOF expansion + conviction upgrade recommendations

6. Build Roadmap — Three Phases to Full FSD

Section 11 expands Phase 1 with the immediate-build frontier specifications. Read Section 11 first, then return here for the full phase context.

Phase	Timeline	Core Deliverables	Milestone
Phase 1	Months 1–2	1. MICS formula — 2-3 days. Highest priority inch. See Section 11.1. 2. Feed-agnostic data ingestion pipeline with free feeds. See Section 11.2. 3. Systematic Scan Engine (~$150/month Claude API). See Section 11.3. 4. Session Log Analytics module. See Section 11.4. 5. Confirmation queue with information-quality response options. See Section 11.5. 6. Brokerage API integration (IBKR/Alpaca). Tier 0 execution live. 7. RPE (Module 5.1) — continuous regime probability scoring.	Tier 0 fully live. MICS replaces PM conviction assignment. Three frontiers closed. Feed receptors built. ARMS begins flying itself.
Phase 2	Months 3–4	1. SENTINEL engine as automated module. Full 6-gate analysis on PM thesis seed. 2. CCM (Module 5.2) — quarterly conviction calibration. Requires 90 days of session log. 3. TRP (Module 5.3) — thesis retirement protocol. 4. AUP (Module 5.5) — asymmetric upside protocol. 5. Model Optimization Engine — quarterly parameter review from session log.	Tier 1 default-execute live. SENTINEL fully automated. Entry and exit equally rigorous. System begins self-calibrating.
Phase 3	Months 5–6	1. PID (Module 5.4) — monthly automated LP report. 2. Full audit infrastructure — immutable session log, regulatory compliance layer. 3. Performance attribution by module. 4. Aggregate Signal Monitor for multi-portfolio and licensing instances. 5. ARMS licensing config layer — fund-agnostic parameter set. ARMS becomes a platform.	Full FSD. Every decision defensible. ARMS is a licensable product. Achelion is a platform.

7. Updated File Structure — v5.0 Target

New files are marked NEW. Modified files are marked UPDATED. Unchanged v4.0 files are marked UNCHANGED.

File	Status	Description
`src/data_feeds/market_data.py`	UPDATED	Feed-agnostic plugin architecture. Free feeds now; institutional feeds plug in later without rebuild.
`src/data_feeds/free_signals.py`	NEW	SEC EDGAR, FRED, USPTO, Quiver Quantitative, Adzuna job data. Phase 1 free signal layer.
`src/engine/regime_probability.py`	NEW	Regime Probability Engine. Every 5 min. Outputs RegimeProbabilitySignal.
`src/engine/macro_compass.py`	UNCHANGED	Core regime detection. Unchanged from v4.0.
`src/engine/aras.py`	UPDATED	Consumes RPE advisory signal. Pre-positioning logic added.
`src/engine/master_engine.py`	UPDATED	Consumes MICS score and CCM calibration factor. C^2 weighting derived from MICS.
`src/engine/mics.py`	NEW	Model-Implied Conviction Score formula. 2-3 days to build. Phase 1 priority #1.
`src/engine/conviction_calibration.py`	NEW	CCM. Runs quarterly. Outputs calibration factor. Phase 2 — needs 90d of log data.
`src/engine/thesis_retirement.py`	NEW	Thesis Retirement Protocol. 3-condition trigger. Queues orderly exit. Phase 2.
`src/engine/asymmetric_upside.py`	NEW	Asymmetric Upside Protocol. 5-condition trigger. Phase 2.
`src/engine/session_log_analytics.py`	NEW	Monthly log analysis. Three output metrics. Phase 1 — learning loop receptor.
`src/engine/systematic_scan.py`	NEW	Weekly SENTINEL scan of AI infrastructure universe. Claude API ~$150/month. Phase 1.
`src/engine/kevlar.py`	UNCHANGED	Position limits. Unchanged from v4.0.

src/engine/perm.py	UPDATED	Connected to TRP for thesis retirement exits.
src/engine/slof.py	UPDATED	AUP expansion eligibility check added.
src/engine/ares.py	UNCHANGED	VARES re-entry sizing. Unchanged from v4.0.
src/engine/drawdown_sentinel.py	UNCHANGED	PDS. Unchanged from v4.0.
src/engine/tail_hedge.py	UPDATED	Consumes RPE pre-positioning signals for PTRH sizing ahead of regime transitions.
src/modules/factor_exposure.py	UNCHANGED	FEM. Unchanged from v4.0.
src/modules/stress_scenarios.py	UNCHANGED	SSL. Unchanged from v4.0.
src/execution/broker_api.py	NEW	Brokerage API (IBKR/Alpaca). Phase 1 priority #2 after MICS.
src/execution/confirmation_queue.py	NEW	Tier 1 veto windows with 3-option information-quality interface. Phase 1.
src/execution/sentinel_engine.py	NEW	SENTINEL v2.0 as automated module. Phase 2.
src/execution/order_book.py	UNCHANGED	LAEP execution. Unchanged from v4.0.
src/reporting/daily_monitor.py	UPDATED	Auto-generates full monitor 6AM CT. All v4.0 + v5.0 module status.
src/reporting/proactive_digest.py	NEW	PID monthly automated LP report. Phase 3.
src/reporting/audit_log.py	NEW	Immutable session log. Every action timestamped with module source. Phase 1.
src/reporting/session_log_analytics.py	NEW	Monthly learning loop metrics. Phase 1.
src/reporting/performance_attribution.py	NEW	Module-level attribution. Phase 3.
src/api/dashboard_api.py	UPDATED	Exposes all new module outputs including MICS, RPE, SLA, scan results.

8. Phase 1 Sprint — Josh's First Two Months

Read Section 11 before this section. Three of the five build items below are expanded with full specifications in Section 11.

Phase 1 has two non-negotiable priorities in order: (1) MICS — closes the conviction assignment problem immediately, zero new APIs required; (2) brokerage API integration — closes the gap between signal and execution. Everything else follows.

8.1 Priority Sequence for Phase 1

1. MICS formula (src/engine/mics.py). 2-3 days. Full spec in Section 11.1. Build this first.
2. Feed-agnostic data ingestion pipeline with free signal plugins (src/data_feeds/). 1 week. Full spec in Section 11.2.
3. Systematic Scan Engine (src/engine/systematic_scan.py). 2 weeks. Full spec in Section 11.3. ~$150/month Claude API.
4. Session Log Analytics (src/engine/session_log_analytics.py). 1 day. Full spec in Section 11.4.
5. Confirmation queue with information-quality response options (src/execution/confirmation_queue.py). Full spec in Section 11.5.
6. Brokerage API integration (src/execution/broker_api.py). IBKR via ib_insync. See Section 8.2.
7. Regime Probability Engine (src/engine/regime_probability.py). Module 5.1.

8.2 Brokerage API Integration

Recommended: Interactive Brokers (IBKR) via the ib_insync Python library. Preferred for institutional use — deeper options support for PTRH, better fill quality, cleaner audit trail for regulatory purposes. Alpaca is acceptable as a simpler REST-based alternative for initial testing.

Minimum viable connection for Phase 1

8. Account connection and position snapshot. ARMS reads live portfolio to calculate allocation deltas.
9. Order submission (market and VWAP). LAEP mode determines which type fires.
10. Fill confirmation and logging. Every executed order writes to audit log with triggering module.
11. Options order support. PTRH put positions require options chain access and spread capability.

8.3 The OrderRequest Interface — Freeze Before Connecting

Every ARMS module that triggers a trade produces an OrderRequest. The order book consumes OrderRequest objects regardless of which module produced them. This interface must be defined and frozen before any module connects to execution.

```
@dataclass class OrderRequest:
    ticker: str
    action: Literal['BUY','SELL','BUY_PUT','SELL_CALL']
    quantity: int
    order_type: str              # 'MARKET' | 'VWAP' | 'LIMIT'
    execution_window_min: int    # from LAEP mode
    slippage_budget_bps: float   # from LAEP mode
    priority: int                # 1-4 kill chain order
    triggering_module: str       # 'ARAS'|'PDS'|'PTRH'|'MICS'|etc.
    triggering_signal: str       # human-readable rationale
    tier: int                    # 0=immediate, 1=veto window
    veto_window_hours: float     # 0 for Tier 0
    confirmation_required: bool
```

8.4 Testing Protocol Before Live Connection

12. Paper trading mode. Connect to IBKR paper account. Run full system 30 days. Every order logged and manually verified. No discrepancies before proceeding.
13. Regime transition test. Force Macro Compass composite from RISK_ON through CRASH and back. Verify every module fires in correct sequence at correct ceiling.
14. Kill chain test. Force PDS DELEVERAGE_1 in paper mode. Verify kill chain executes to 60% ceiling with correct priority order. Verify audit log captures every order.

8.5 Day One Checklist for Josh

- Read PM Playbook v4.0 and THB v4.0 in full. Both are canonical. No prior version is relevant.
- Read Section 11 of this document before writing any code.
- Review Architecture AB canonical weights and the five-regime ceiling table.
- Set up IBKR paper account. Obtain API credentials.
- Install ib_insync. Confirm connection to paper account from dev environment.
- Read the existing ARMS codebase zip files from MJ. Map every module to the Layer diagram in Section 1.
- Define and freeze the OrderRequest interface (Section 8.3) before connecting anything to execution.
- Build src/engine/mics.py first. Two to three days. This is the highest-value inch available.

9. The LP Narrative

"Most funds tell you they have a disciplined process. We have removed the human from the process entirely — except where human judgment is genuinely irreplaceable."

What most funds say	We have a disciplined investment process managed by an experienced team.
What Achelion says	We have removed the human from the execution process entirely. Our PM does one thing no algorithm can replicate: he identifies what the world will look like before the data confirms it. Everything after that is engineering.
What this means for LPs	You are not betting on our judgment in a crisis. You are betting on an architecture that was designed specifically for the crisis you have not seen yet.
How we prove it	The session log is immutable. Every decision has a timestamp, a triggering signal, and the module that generated it. You can verify the system ran exactly as described in every single session.
The Buffett correction	Far more money has been lost on Wall Street by salesmen than by investors. ARMS removes the curtain entirely. Judge us on the process, not the narrative.
On MICS specifically	Our conviction scoring is not assigned by a portfolio manager. It is derived from a quantitative formula applied to the same gate data that governs every entry decision. The PM can adjust it by one level with a documented reason. That is the entire scope of human involvement in sizing.

10. Closing Statement

The factory is built. The robots are ready. The humans are in the control room.

What follows from the completion of this build is not an incremental improvement to how Achelion manages capital. It is a categorical change in what a hedge fund can be.

For the first time at the emerging fund level, a portfolio will be managed by a system that detects regime transitions before they happen, scores conviction from data rather than opinion, executes without emotional interference, and gets smarter over time through calibration and thesis retirement feedback loops — while a human partner brings the one thing no system can replicate: the ability to see the future before it is in the data.

The best version of human judgment is not a human trying to outperform a machine in the machine's domain. It is a human working with a machine in a partnership where each does what it does best — and neither pretends to do the other's job.

"Silence is trust in the architecture."

Achelion Capital Management, LLC · Flow. Illumination. Discipline. Conviction.

11. Immediate-Build Frontiers — What Josh Builds Now

Inches today, miles later. Every receptor built now accepts the expensive upgrade without rearchitecting. This section is Josh's first read.

Of the five long-term frontiers identified in the strategic architecture review, three are fully buildable today at zero or near-zero incremental cost. Two require expensive data feeds or infrastructure not yet justified by current AUM — but their receptors must be built now so the upgrades are configuration changes, not rebuilds.

Frontier	Build now?	Incremental cost	Section
F3 — Model-Implied Conviction Score (MICS)	YES — fully close	$0	11.1
F1 — Proprietary signal layer (receptors)	YES — receptors only	$0 (free APIs)	11.2
F2 — Autonomous thesis generation (low-fidelity)	YES — low fidelity now	~$150/month	11.3
F4 — Learning loop (receptors)	YES — receptors only	$0	11.4
F5 — Information-quality confirmation windows	YES — design decision now	$0 (design choice)	11.5
F1 — Institutional data feeds (full)	Phase 4 — AUM dependent	$50K–$500K/year/feed	Future
F2 — High-fidelity autonomous thesis generation	Phase 4 — AUM dependent	Institutional alt data	Future

11.1 Model-Implied Conviction Score (MICS) — Build Time: 2-3 Days

Highest-value inch. Closes Frontier 3 almost completely. Zero new APIs. Zero ongoing cost.

MICS replaces the PM's free conviction level assignment with a scored output derived entirely from SENTINEL gate data. The formula is pure Python arithmetic on data SENTINEL already produces. The PM's role becomes a bounded confirmation — not a free assignment.

The formula

```
# Component weights — sum to 1.0
gate3_weight   = 0.40  # Quantitative mispricing — core signal
gate6_weight   = 0.30  # Source quality — information edge
gate4_weight   = 0.15  # FEM impact — portfolio cleanliness
gate5_weight   = 0.15  # Regime timing — entry quality

# Gate 3: normalize raw score (0-30) to 0-10
g3_score = (gate3_raw / 30) * 10

# Gate 6: source category -> score
source_scores = {'Cat A': 10, 'Cat B': 8, 'Cat C': 5, 'None': 4}
g6_score = source_scores[source_category]

# Gate 4: FEM impact -> score
fem_scores = {
    'NORMAL->NORMAL':          10,
    'NORMAL->WATCH':             7,
    'WATCH->WATCH':              6,
    'WATCH->ALERT (paired trim)':4
}
g4_score = fem_scores[fem_impact]

# Gate 5: regime at entry -> score
regime_scores = {
    'RISK_ON':          10,
    'WATCH':             9,
    'NEUTRAL':           7,
    'NEUTRAL (queued)':  5
}
g5_score = regime_scores[regime_at_entry]

# MICS raw (0.0 - 10.0 float)
MICS_raw = (g3_score * gate3_weight) +
           (g6_score * gate6_weight) +
           (g4_score * gate4_weight) +
           (g5_score * gate5_weight)

# C-level (integer 1-10)
MICS = max(1, min(10, round(MICS_raw)))
```

PM override — bounded, not free

PM action	Condition	Process	Audit
Accept MICS score	Default	Silence or one-tap confirm	Auto-logged

Adjust ±1 level	PM has specific documented rationale	Written reason in Gate 6 form	Permanent
Adjust ±2 or more	Extraordinary circumstance only	GP co-sign required + written rationale	Permanent + GP flag
Category D attempt	Not permitted	System rejects. No input field exists.	Rejection logged

Retroactive scoring of current book

After MICS is built, run every current Architecture AB position through the formula using best available proxy data for the original SENTINEL gate inputs. Where gate data is unavailable, use current Gate 3 scores and current source category estimates. Any position where MICS diverges from the current PM-assigned C-level by more than 2 levels is flagged for GP review — not to force a change, but to surface the divergence and require a documented explanation.

File	src/engine/mics.py
Build time	2-3 days
Dependencies	SENTINEL gate outputs only — no new APIs
Ongoing cost	$0
Mile later	CCM feeds quarterly calibration factor into MICS weights. Scoring improves automatically from its own track record.

11.2 Feed-Agnostic Data Ingestion Pipeline — Build Time: 1 Week

Build the receptor now. The expensive feeds plug in later as configuration changes, not code rebuilds.

Every data feed must be implemented as a plugin conforming to a standard FeedPlugin interface. The pipeline does not care whether a feed comes from a free API or a $500K institutional subscription — it only sees the interface. When alternative data becomes affordable, it is a new plugin, not a new architecture.

FeedPlugin interface

```
class FeedPlugin(ABC):
    @abstractmethod
    def fetch(self) -> List[SignalRecord]: ...
    # Every feed returns List[SignalRecord]
    # regardless of source or cost tier


@dataclass class SignalRecord:
    ticker: str          # or 'MACRO' for macro signals
```

```
signal_type: str     # e.g., 'EARNINGS_VELOCITY'
value: float         # normalized 0-1 where applicable
raw_value: Any       # original value before normalization
source: str          # feed name e.g., 'SEC_EDGAR'
timestamp: str       # ISO format
cost_tier: str       # 'FREE' | 'LOW' | 'INSTITUTIONAL'
```

Free feeds to connect in Phase 1

Feed	Signal type	API	Cost	Mile later
SEC EDGAR full-text	Earnings language velocity, 13F position changes, insider filings	api.sec.gov — free	$0	Replace with institutional 13F analytics
FRED (Federal Reserve)	Macro regime inputs: yields, CPI, PCE, credit spreads, PMI	api.stlouisfed.org — free	$0	Already institutional quality — no upgrade needed
USPTO Patent Bulk Data	Patent filing velocity by company and technology category	patentsview.org/api — free	$0	Replace with specialized patent analytics
Quiver Quantitative	Congressional trading disclosures, dark pool flow, institutional ownership	quiverquant.com — free tier	$0	Upgrade to paid tier at $50/month for full flow
Adzuna Job Postings API	Job posting velocity as capex leading indicator	api.adzuna.com — free tier	$0	Replace with Revelio Labs or similar at scale
MarineTraffic free tier	Shipping activity as supply chain proxy	marinetraffic.com — free tier	$0	Replace with Orbital Insight satellite data

File	src/data_feeds/free_signals.py + individual plugin files
Build time	1 week
Ongoing cost	$0
Mile later	Each free feed is replaced by an institutional equivalent by swapping one plugin. The pipeline, ARAS, and all downstream modules require zero changes.

11.3 Systematic Scan Engine — Build Time: 2 Weeks · ~$150/month

Low-fidelity autonomous thesis generation. PM shifts from finding ideas to evaluating system candidates.

The Scan Engine runs every Monday pre-market against a defined universe of AI infrastructure companies. It applies SENTINEL Gates 1 and 2 automatically using Claude API analysis of publicly available earnings transcripts, SEC filings, and patent data. Any company passing both gates receives a Gate 3 score from public valuation data. Results are delivered as a weekly intelligence briefing — not as trade recommendations, but as candidates the PM has not yet examined.

Process

15. Pull the latest 10-K/10-Q/8-K filing and most recent earnings call transcript for each universe company from SEC EDGAR (free).
16. Send filing and transcript to Claude API with a structured SENTINEL Gate 1 and Gate 2 prompt. Request JSON output: {gate1_pass: bool, gate2_pass: bool, gate1_rationale: str, gate2_rationale: str}.
17. For companies passing both gates, compute Gate 3 score from public valuation data (P/E vs peers, institutional positioning from 13F, analyst consensus framing gap).
18. Generate weekly briefing: companies passing Gates 1-2 with Gate 3 scores, ranked by mispricing score. Deliver to PM as Section 3 of Monday daily monitor.
19. PM reviews as intelligence — not as instructions. If a candidate matches pre-consensus intelligence, PM initiates a full SENTINEL run.

Universe definition

Start with 50-75 companies across AI infrastructure: semiconductors (fabless and foundry-dependent), networking, power/cooling infrastructure, data center REITs, cloud platforms, AI software with infrastructure exposure. Add or remove tickers via a config file — no code changes required.

Cost estimation

Item	Volume	Unit cost	Monthly cost
Claude API — Gate 1+2 analysis	75 companies × 4 weeks = 300 calls	~$0.40/call average	~$120
Claude API — Gate 3 scoring	~40 pass rate × 4 weeks = 160 calls	~$0.20/call	~$32
SEC EDGAR filing pull	API calls for transcripts and filings	$0 — free API	$0
Total			~$150/month

File	src/engine/systematic_scan.py
Runs	Every Monday 5AM CT

Build time	2 weeks
Ongoing cost	~$150/month Claude API
Mile later	Replace Claude API analysis with fine-tuned model on ARMS proprietary thesis log. Replace free filing pulls with institutional transcript service. Gate 3 scoring upgrades automatically as free signal feeds are replaced with institutional alternatives.

11.4 Session Log Analytics (SLA) — Build Time: 1 Day

Three numbers monthly. The learning loop receptor. Zero cost. Feeds the Model Optimization Engine in Phase 2.

SLA reads the session log ARMS already writes and outputs three metrics monthly. These metrics are the foundation of the learning loop — when the CCM and Model Optimization Engine are built in Phase 2, they read from this same log. Build the structured log correctly now and the learning loop has everything it needs when the time comes.

Metric	Definition	Why it matters
CDF accuracy rate	% of positions that hit day-45 decay and subsequently exited at a loss vs. recovered. Measures whether the 45-day trigger is calibrated correctly.	If >70% of decayed positions eventually exit at a loss, the trigger is right. If <40%, the 45-day window may be too short.
Regime transition lag	Average days between first signal elevation and formal regime call across all transitions in the period. Measures RPE pre-positioning effectiveness.	As RPE matures, this lag should shrink. Tracks whether pre-positioning is working.
SENTINEL Gate 3 predictive accuracy	Spearman rank correlation between Gate 3 scores at entry and subsequent 90-day performance vs QQQ. Measures whether the mispricing score actually predicts outperformance.	Feeds directly into CCM and MICS weight optimization in Phase 2. The most important long-run metric in the system.

Session log structure — build this correctly from day one

Every system action must write to the session log with this minimum structure. If the log is unstructured, SLA cannot read it and the learning loop has no data to consume.

```
@dataclass class SessionLogEntry:
    timestamp: str            # ISO format — millisecond precision
    action_type: str          # 'REGIME_CHANGE'|'TRADE'|'CDF_DECAY'|etc.
    triggering_module: str    # 'ARAS'|'PDS'|'MICS'|'CDF'|etc.
    triggering_signal: str    # human-readable rationale
```

```
    ticker: Optional[str]     # None for portfolio-level actions
    mics_score: Optional[float]  # at time of action
    regime_at_action: str     # canonical regime label
    gate3_score: Optional[float]  # for SENTINEL actions
    source_category: Optional[str]  # for SENTINEL actions
    pm_override: bool         # True if PM deviated from system rec
    override_rationale: Optional[str]
    outcome_90d: Optional[float]  # filled in 90 days after entry
```

File	src/reporting/audit_log.py + src/engine/session_log_analytics.py
Build time	1 day (log structure + 3 metric calculations)
Ongoing cost	$0
Mile later	CCM reads outcome_90d field. Model Optimization Engine reads all fields. Performance attribution by module reads triggering_module. Everything flows from this log.

11.5 Information-Quality Confirmation Queue — Design Decision Now

This costs nothing extra to implement correctly at build time. It costs weeks to retrofit later. Make this design decision today.

The current specification describes time-based veto windows (4 hours, 2 hours, 24 hours). This is correct as a fallback. But the primary interface must be built around information quality, not elapsed time. When the PM has nothing to add, they should be able to execute immediately — not wait out a 4-hour clock. When they have Category A intelligence, they need as much time as they need. The design decision is made now, at zero extra cost, because redesigning the interface in Phase 2 after LPs have seen it is avoidable.

Three-option response interface — replace time-first with information-first

Response option	Meaning	System action	Audit
Option 1: No new information — execute	PM has reviewed the system recommendation and has nothing to add. Silence is trust in the architecture.	Executes immediately at MICS-implied sizing. No clock delay.	Logged as 'PM confirmed — no new information'
Option 2: New information present — hold	PM has Cat A or B intelligence that may change the recommendation. Opens source declaration form.	Pauses execution. PM declares source category and specific insight. System re-runs Gate 3 with updated source score if Cat A/B is	Full source declaration logged permanently

		declared. MICS updates. Execution fires with updated score.	
Option 3: Veto with rationale	PM disagrees with the system recommendation and is blocking execution.	No execution. PM must write a specific rationale. Logged permanently. If PM vetoes more than 3 Tier 1 recommendations in 30 days, GP review triggered automatically.	Permanent log + frequency monitoring
No response (time-based fallback)	PM did not respond within the defined window. Clock-based default.	Executes at MICS-implied sizing. Identical to Option 1. Time window is the fallback for PM unavailability, not the primary interface.	Logged as 'PM non-response — system executed'

Veto frequency monitoring — the override audit

If the PM vetoes more than 3 Tier 1 recommendations in any 30-day period, the system automatically generates a veto frequency report for GP review. This is not punitive — it is informational. It surfaces whether the PM and the system are systematically disagreeing, which is itself a signal. Either the model needs recalibration or the PM's judgment is drifting from the architecture. Both are important to know.

File	src/execution/confirmation_queue.py
Build time	Part of Phase 1 brokerage integration sprint
Ongoing cost	$0
Mile later	When AUM supports it, the confirmation queue integrates with a mobile app for push delivery and one-tap response. The interface design does not change — only the delivery mechanism.

11.6 Summary — Total Cost of Immediate-Build Frontiers

Build item	Build time	Ongoing cost	Frontier closed	Mile it enables

MICS formula (mics.py)	2-3 days	$0	F3 — fully	CCM weight optimization in Phase 2
Feed-agnostic pipeline + free feeds	1 week	$0	F1 — receptor	Institutional feeds plug in as config changes
Systematic Scan Engine	2 weeks	~$150/month	F2 — low fidelity	Fine-tuned model replaces Claude API at scale
Session Log Analytics	1 day	$0	F4 — receptor	CCM + Model Opt Engine read from this log
Information-quality confirmation queue	Part of Phase 1	$0	F5 — fully	Mobile app delivery at scale; interface unchanged
TOTAL	**~4-5 weeks**	**~$150/month**	**3 fully closed 2 receptors built**	**All 5 frontiers accessible at Phase 4**

The total capital required to move all five frontiers from future aspiration to active receptor or full implementation is approximately $150 per month. The total build time before any of this is in production is four to five weeks for Josh working in parallel with Phase 1. There is no justification for waiting. These are the inches. Build them.

"Silence is trust in the architecture."

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